the same trading trajectory for certain parameter values. This was shown by Roberto Malamut (Ph.D., Cornell University) via simulation techniques. In addition, even if the estimated parameters are close to true parameter values for both models the resulting trading trajectories from each model will be extremely close to one another, thus resulting in the same underlying trading strategy. We, however, have found an easier time finding a relationship between cost and order size using the I-Star impact model. But we encourage readers to experiment with both approaches to determine the modeling technique that works best for their needs. Calibration of the I-Star model and required data set is described next.

Underlying Data Set

The underlying data set needed to fit the I-Star model shown above includes: $Q = imbalance\ or\ order\ size$, $ADV = average\ daily\ volume$, $V^* = actual\ trading\ volume$, $\sigma = price\ volatility$, $POV = percentage\ of\ volume$, and $Cost = arrival\ cost$. These variables are described as follows.

Imbalance/Order Size

Recall, we based the I-Star model on total market imbalance (e.g., differences in buyer and seller transaction pressure). Regrettably, exact buy-sell imbalance at any point is neither known nor reported by any source so we have to infer this information from various data sets and techniques. The following methodologies have been used to infer order sizes and imbalances:

Lee and Ready Tick Rule—imbalance is defined as the difference between buy-initiated and sell-initiated trades in the trading interval. A positive imbalance signifies a buy-initiated order and a negative imbalance signifies a sell-initiated order. The Lee and Ready tick rule maps each trade to the market quote at that point in time. Trades at prices higher than the bid-ask spread mid-point are denoted as buy-initiated and trades at prices lower than the bid-ask spread mid-point are designated as sell initiated-trades. Trades exactly at the mid-point are designated based on the previous price change. If the previous change was an up tic, we designate the trade buy-initiated; down tic, a sell-initiated trade. The modified Lee and Ready tick rule assigns trades as buy- or sell-initiated based on price change. An up tic or zero-up tic is known as a buy-initiated trade and a down tic or zero-down tic represents a sell-initiated trade. The proliferation of dark pools prompted many practitioners to exclude trades that occurred inside the bid-ask spread from being designed as buy- or sell-initiated. The difference between buy-initiated and sell-initiated trades is denoted as the order